Human-Computer Interaction: Design & Evaluation

Lab 1 exercise sheet – Week 2 (Friday 23rd, January 2026)

This week's lab work is not assessed. There is nothing to submit.

At this point, you should have identified your team project group by finding it in the Self-Grouping activity on Moodle. If you don't know your group yet, or somehow haven't been assigned to one, your priority in the lab is to do so. As your first task, please find your group members, and inform your tutors/lab assistants if you haven't been assigned to a group or found your group yet. In this first lab, you will get to meet your group members and get used to working together with each other on two example HCI-inspired experiences.

We have some warm-up tasks below to get you thinking about some of the HCI issues that will be covered in the course.

Typing

In this week's lab, we'll be reproducing a typing experiment conducted by a research team from Aalto university (Palin, K., Feit, A., Kim, S., Kristensson, P., O., & Oulasvirta, A. 2019. *How do People Type on Mobile Devices? Observations from a Study with 37,000 Volunteers.* MOBILEHCI'19). They collected a large-scale data set recorded using a typing test of their own creation.

With your lab group, try a typing test for yourselves, first on a laptop/desktop, then on a mobile phone, and take a note of your **WPM** (words per minute) and **Error** results. Everyone in your lab group of 5-6 people should do it individually, then collect your results together as a group. There are many online typing tests available, we suggest using https://www.typingtest.com/ as it was the website used in Palin et al's study. (Consider alternatives such as https://www.arealme.com/typing-test/en/ if it is unavailable).

Compare your results with your group. For instance, consider the following:

- Did everyone get a similar performance on **WPM** and **Error**? Try plotting the results using a simple graph (eg creating a graph with Excel).
- How do **WPM** and **Error** metrics vary together? Do classmates with a higher **WPM** also get few errors, or do they get more errors?
- How did it feel to do the test between desktop/laptop and mobile phone: is it easier on one device? Do the WPM and Error provide a good representation of how it feels to type with that device, or is something missing?

Here are some selected conclusions from Palin et al: do your results seem to agree with their findings?

1. With only 1 or 2 fingers, people type about 70% as fast on mobile devices as on full desktop keyboards. Still, the average performance is only around 36 WPM.
2. People using auto-correction type faster. In contrast, people who manually chose words suggested by the keyboard type slower.

In HCIDE, we will discuss many of these topics: from differences in performances across people or devices, and how to design better devices and interfaces and how to evaluate them for their ease of use, or usability.

Learning and User Expectations

Users expect certain conventions to be followed, and when these break down user interfaces can become challenging. As developers, we are often blinded to errors in our own work because we have "learned" how to use our own interfaces.

The following online experience "User Inyerface" (https://userinyerface.com) tries to demonstrate this by creating purposefully poorly designed UI. It does an excellent job of pointing out how important interface conventions are, but also how with a small amount of training users can adapt.

Try to complete the User Inyerface challenge one time, recording how quickly you can complete all the tasks on your first run through. Again, compare your results as a group.

Then, go through the challenge again but instead of attempting to beat it as fast as possible, try to identify as many issues as you can that makes the experience difficult to use. Write each issue them down, giving each issue a short descriptive title.

Once done, analyse the issues that you identified. Can you group them into major categories of UI problems?

You do not have to submit this work anywhere.